TEXORA · COURSE NOTES · LECTURE SHEET

Communication Network

The patterns by which messages travel in a group — chain, wheel, circle, Y and all-channel networks, and what each pattern does to speed, accuracy, leadership and morale.

HUM-1107 · COMMUNICATIVE ENGLISH · TOPIC 6 · LEVEL 1 TERM 1

1. What a network is

Put five people to work on a shared problem and the question is no longer whether they communicate, but **through what pattern**: who may speak to whom, whose desk every message must cross, who is isolated at a rim and who sits at a hub. That pattern is the group's **communication network**, and laboratory studies of small groups (the classic experiments of Bavelas and Leavitt) showed that the pattern alone — same people, same task — changes speed, accuracy, who becomes the leader, and how satisfied the members feel. Organisations design networks deliberately (organisation charts are network diagrams) and grow them accidentally (the grapevine is a network nobody drew).

2. The five classic patterns

NETWORK	STRUCTURE	BEHAVIOUR
Chain	Each person communicates only with the one above and the one below — the line of command drawn literally	Orderly and accurate while the chain is short; slow, and distortion compounds link by link (Topic 2's organisational barrier, diagrammed). No single leader emerges; end members feel isolated.
Wheel (star)	All members communicate only through one central person, who talks to everyone	Fastest for simple problems and very accurate — with the hub as unmistakable leader. The rim members have the lowest satisfaction; the hub is a bottleneck and a single point of failure.
Circle	Each person communicates with the two neighbours on either side	No leader, high member satisfaction and good morale; but slow and prone to drift — no one holds the whole picture, so organisation is poor.

Y	A chain with a fork: two branches feeding one person, who reports upward	Between chain and wheel: a gatekeeper position forms where the fork joins — the person there gains influence by filtering what passes (the secretary, the shift in-charge).
All-channel (completely connected)	Every member communicates freely with every other	Best for complex, creative problems and highest overall satisfaction; leadership is shared. Slowest to converge on simple tasks and can dissolve into noise without discipline.

3. What the experiments teach

- **Centralised networks (wheel, chain, Y)** solve simple, routine problems faster and with fewer errors — information converges on one desk that decides.
- **Decentralised networks (circle, all-channel)** outperform on complex, ambiguous problems, because information pools where it is needed instead of queueing at a hub — and their members report far higher satisfaction and commitment.
- **Leadership follows position:** the occupant of the most central position is perceived as the leader regardless of ability — a warning against mistaking the in-tray for the talent.
- **Morale is inverse to centrality for the rim:** peripheral members of centralised networks disengage — the organisational cost that eventually eats the speed advantage.

4. Formal and informal networks in the real organisation

The formal network is the chart: chain-like along departments with wheel-like hubs at each manager. Overlaid on it, invisible and undrawn, sits the informal network — clusters of friendship and expertise through which the grapevine runs (Topic 3), usually nearer to all-channel within each cluster and to a chain of trusted links between clusters. Effective organisations read both maps: they shorten formal chains that distort (delayering), create all-channel pockets for problem-solving (project teams, quality circles), and never pretend the informal map does not exist. In the digital mill the patterns persist in new dress — the email cc-list is a chosen network, the messaging group a wheel around its admin or an all-channel room, and the same design questions apply.

5. Glossary — key terms of this topic

TERM	MEANING
Communication network	The patterned set of channels linking the members of a group or organisation.

Centralisation	The degree to which communication is concentrated through one position.
Gatekeeper	A person positioned where flows join, able to filter what passes (the Y-network's fork).
Hub / rim	The central position of a wheel network / its peripheral positions.
Quality circle	A small voluntary group meeting in an all-channel pattern to solve work problems.
Delayering	Removing levels from an organisation's chain to shorten communication paths.

6. Key points and summary

- A network is who can talk to whom; the five classic patterns — chain, wheel, circle, Y, all-channel — trade speed and accuracy against satisfaction and adaptability.
- Centralised patterns win simple tasks and create leaders by position; decentralised patterns win complex tasks and morale.
- Real organisations overlay a designed formal network with an undrawn informal one; managing both maps is a core management skill.

7. Practice

Answer in your own words; brief model answers follow.

1. Sketch (in words) the five classic networks and state which single structural feature principally distinguishes them.
2. A five-person team must solve (a) a simple tallying task against the clock, (b) an open-ended product problem. Choose a network for each and justify by the experimental findings.
3. In a wheel network, why does a leader "emerge" even if the hub person is not the ablest member?
4. A message crossing six links of a chain arrives distorted. Name the barrier type and the structural remedy.
5. Where is the gatekeeper in a Y network, and why does the position carry power?
6. Map a garment factory's quality department onto network language: inspectors report to a quality in-charge, who alone reports to the quality manager, who alone speaks to production management. Name the pattern(s) and predict the morale and speed effects.
7. Why do messaging-app groups behave sometimes like wheels and sometimes like all-channel networks? Give the deciding factor.

Answer key

1. Chain: a line, each linked to neighbours only. Wheel: spokes to one hub. Circle: a closed ring of neighbour links. Y: a chain forked at one end. All-channel: every pair linked. The distinguishing feature is the pattern of permitted channels — equivalently, how centralised the pattern is.
2. (a) Wheel (or chain): centralised patterns are fastest and most accurate on simple tasks. (b) All-channel: decentralised patterns pool information for complex problems and keep members engaged.
3. Leadership follows centrality: all information passes through the hub, so the hub sees the whole problem first and issues the organising messages — the group reads the position's behaviour as leadership.
4. An organisational barrier — distortion accumulated over relay links. Remedies: shorten the chain (delayer), add feedback at each link, or put the message in writing so it relays unchanged.
5. At the fork, where two branches join into one upward channel. Everything from both branches reaches management only through that desk, which can filter, delay or emphasise — information control is power.
6. A chain topped by a wheel-hub at each level — effectively nested wheels in a chain. Prediction: fast, accurate upward reporting of simple counts, a clear leader at each hub, but inspectors at the rim report low involvement and their local knowledge reaches decisions slowly, if at all.
7. Whether contributions actually flow member-to-member or only member-to-admin-to-members: an announcement group where only the admin posts is a wheel; an open discussion group is all-channel. The deciding factor is the permitted-channel pattern, not the app.
